

Difference-in-Differences

Staggered Adoption: Three Parallel-Trends Assumptions, Three Estimators

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Sum up, and what comes next

The previous chapter had **one treatment date**: a treated group, a control group, before and after. The event study put one coefficient per period; parallel trends was one assumption.

Most policies occur in stages. They **roll out**: some units adopt early, some late, some never. Unilateral divorce laws by US state, minimum wages by state, broadband by municipality, emission-control areas by sea, organic certification by farm.

This chapter

Under staggered adoption the two-way fixed-effects regression mixes clean comparisons with comparisons to already-treated units, and its coefficient can be biased by treatment dynamics alone. This chapter shows why, and replaces the regression by an explicit construction: group-time effects identified under a stated parallel-trends assumption, then aggregated with transparent weights.

Reference for the whole chapter: Baker, Callaway, Cunningham, Goodman-Bacon & Sant'Anna (2026), "Difference-in-Differences Designs: A Practitioner's Guide".

Roadmap

- 1 **The setting** — cohorts, event time, never- and not-yet-treated units.
- 2 **What TWFE computes** — forbidden comparisons, negative weights, contaminated leads and lags.
- 3 **Identification: the choice of comparison group and base period** — the parameter $ATT(g, t)$; three parallel-trends assumptions (never-treated, not-yet-treated, all periods); each estimator as the sample analogue of one of them.
- 4 **Aggregation and workflow.**

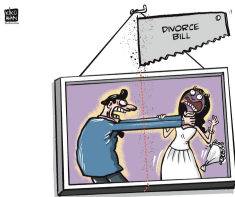
Section 1

Setting

Running example: unilateral divorce in the US

Until the late 1960s a US divorce required both spouses' consent, or proof of fault. **Unilateral divorce** laws let one spouse end the marriage alone: California first (1970), most states in the 1970s, some never.

- **Why it matters.** The **exit option** changes bargaining power inside the marriage. Stevenson and Wolfers (2006) ask whether it reduced women's distress: **women's suicide rates**, domestic violence, spousal homicide, by state and year.
- **Why it is a staggered design.** One adoption year per state, some states never adopt, effects that build up over years. Goodman-Bacon (2021) re-analysed it; we return to it at each step.



Cohorts, event time, comparison groups

To write the divorce example in mathematics, consider a unit i (a state) first treated at date g_i , its **cohort**; never-treated units have $g_i = \infty$. **Event time** is $k = t - g_i$, the number of years since adoption.

- At date t a unit is **treated** ($t \geq g_i$), **not yet treated** ($t < g_i < \infty$) or **never treated**. In 1976, a state that adopted in 1973 is treated ($k = 3$); a state adopting in 1977 is not yet treated; a state that never adopts is never treated.
- The last two are the candidate **comparison groups**: the chapter is about which of them to use, and from which base year.

Section 2

What TWFE computes

Two regressions, and the plan of the chapter

With a treatment date per unit, two regressions come to mind: the **static** TWFE of chapter 4,

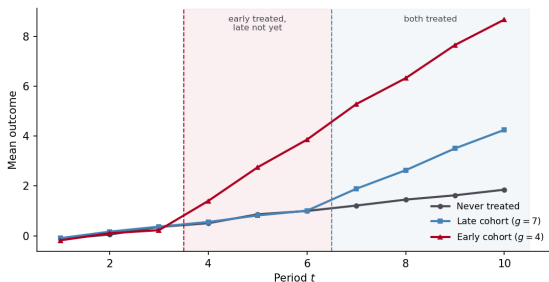
$$y_{it} = f_i + f_t + \beta d_{it} + u_{it}, \quad d_{it} = \mathbf{1}\{t \geq g_i\},$$

and its **dynamic** version, one coefficient per event time,

$$y_{it} = f_i + f_t + \sum_{k \neq -1} \beta_k \mathbf{1}\{t - g_i = k\} + u_{it}.$$

The next slides show what β and the β_k compute under staggered timing. The rest of the chapter replaces them by an explicit construction: one effect per cohort and date, identified under a stated parallel-trends assumption, then aggregated.

A simulated roll-out



Three cohorts (early- and late-adopting states, never-adopting states); effects **grow** with time since treatment and differ across cohorts. In periods 4–6 the late cohort is a clean comparison for the early one. In periods 7–10 TWFE still compares the late cohort to the early one, already treated and still responding: its change is not a counterfactual, a **forbidden comparison**. That change contains the early cohort's continuing effect, and TWFE subtracts it: this is what a **negative weight** on a treatment effect means.

What TWFE computes under staggered adoption

With many periods and cohorts, the static TWFE coefficient is a weighted average of **all** the 2×2 DiDs in the data (Goodman-Bacon 2021),

$$\beta_{\text{TWFE}} = \sum_j w_j \text{DiD}_j, \quad \sum_j w_j = 1 :$$

early vs never, late vs never, late vs early before the late adopts (clean), **early vs late** after the late adopts (forbidden).

- The weights w_j are positive and mechanical (group sizes, timing), unrelated to the policy question.
- In a forbidden term the control's change contains its own effect, which is subtracted: some cohorts' treatment effects get **negative weights** (de Chaisemartin and D'Haultfoeulle 2020); the coefficient can shrink or change sign even when every effect is positive (compare compliers in IV, Rotemberg weights in shift-share).
- Running example (Goodman-Bacon 2021): the static TWFE gives -3.1 suicides per million women; without the forbidden comparisons the effect is higher and reaches -5.4 .

Section 3

Dynamic TWFE

The event study: contaminated leads and lags

The dynamic regression, one coefficient β_k per event time, has the same problem. Sun and Abraham (2021) established it:

What the dynamic TWFE coefficients compute

Under staggered timing, each $\hat{\beta}_k$ is a weighted average of the cohorts' own effects k periods after adoption *and* of effects at **other** event times from **other** cohorts — with weights that can be negative. The leads ($k < 0$) of the late cohorts are measured against early cohorts whose treatment effects are still building up, so **the early cohorts' effects appear in the late cohorts' pre-period coefficients**.

- The contamination comes from the same source: cohorts serve as each other's controls while they are themselves responding.
- It vanishes only if effects are homogeneous across cohorts — precisely what an event study is meant to explore.

Section 4

Three assumptions

The solution: one 2×2 per cohort and date

The solution to the problem just shown: estimate the effect of each cohort at each date from a comparison chosen explicitly, not left to the regression. This starts from a definition of the effect for one cohort at one date. Potential outcomes carry the adoption date: $Y_{it}(g)$ if first treated at g , $Y_{it}(\infty)$ if never; no anticipation: $Y_{it}(g) = Y_{it}(\infty)$ for $t < g$.

$$\text{ATT}(g, t) = \mathbb{E}[Y_{it}(g) - Y_{it}(\infty) \mid g_i = g], \quad t \geq g,$$

the effect at date t for the cohort first treated at g : a **group-specific event study**, one effect per cohort and date.

Forward engineering (Baker et al. 2026)

Every $\text{ATT}(g, t)$ is identified by a 2×2 : cohort g against a **comparison group**, period t against a **base period**. A staggered design is a collection of such 2×2 s, then an aggregation.

Two choices remain, *which comparison group* and *which base period*; both are set by the parallel-trends assumption one is willing to make (next slide).

Three parallel-trends assumptions

	Untreated trends equal for...	Comparison	Base period
PT-never	cohort g and the never-treated , post-periods only	never-treated	$g - 1$ only
PT-not-yet	cohort g and any unit not yet treated at t , post-periods only	never- and not-yet-treated	$g - 1$ only
PT-all	every pair of cohorts in every period, pre and post	any not-yet-treated	any pre-period (or their average)

Running example: the states that adopted in 1973, observed in 1976. PT-never compares them with the never-adopting states only; PT-not-yet adds the states adopting in 1977 or later; PT-all also uses the years before 1972 as base periods.

- Under PT-never or PT-not-yet, parallel trends is imposed **only after** treatment; the only justifiable base period is $g - 1$. Pre-trends are *evidence*, not part of the assumption.
- Under PT-all, parallel **pre-trends** are part of the assumption — which buys precision and makes pre-trends **testable** (more restrictions than parameters: the model is over-identified).

Pre-trends in staggered designs

- **Low power:** joint tests on the leads rarely detect violations of a magnitude that matters (Roth 2022); visual inspection of the leads is not sufficient evidence.
- Report **uniform confidence bands** for the whole event-study curve: a band that contains the entire true curve with 95 % probability, wider than pointwise intervals (the `did` package does by default).
- **Sensitivity:** Rambachan and Roth (2023) bound the effect under violations no larger than the largest pre-period deviation, which turns the inspection of pre-trends into a quantitative statement.
- Cohorts that plausibly **selected on recent outcomes** violate every version: remove them and say so, rather than let them contaminate the comparison.

Estimators are sample analogues of an assumption

Estimator	Assumption	What it is
Callaway & Sant’Anna (2021)	PT-never <i>or</i> PT-not-yet	the plug-in (sample means in place of expectations) $[\bar{y}_{g,t} - \bar{y}_{g,g-1}] - [\bar{y}_{C,t} - \bar{y}_{C,g-1}]$, C = never- or not-yet-treated
Sun & Abraham (2021)	PT-never	the same never-treated estimand, from a regression with one dummy per cohort and event time
Wooldridge (2021); Gardner (2022) did2s; Borusyak et al. (2024)	PT-all	imputation : fit unit and time effects on untreated cells, average $y - \hat{y}(0)$ over treated cells; identical to each other in balanced panels

Sun–Abraham and Callaway–Sant’Anna (never-treated): **one estimand, two computations**. Imputation and Sun–Abraham: **two estimands** — they need not agree.

Other constructions: stacked regressions (Cengiz et al. 2019), one clean 2×2 per cohort; de Chaisemartin and D’Haultfœuille (2020), switchers against not-yet-switchers, which also allows the treatment to switch off (next chapter).

Which assumption? The real trade-off

PT-all (imputation)

- uses *all* pre-periods and *all* not-yet-treated units $\Rightarrow$ the most **precise**;
- but imposes parallel *pre*-trends: if they fail, the estimate is **biased**;
- over-identified $\Rightarrow$ pre-trends become a genuine **test**.

PT-never (Sun–Abraham; Callaway–Sant’Anna with the never-treated)

- comparison group of **fixed composition**; no restriction on pre-trends;
- never uses a unit that chose to adopt *because* of its recent outcomes;
- but never-treated units may be **too different**, or **too few**.

PT-not-yet (Callaway–Sant’Anna with the not-yet-treated; de Chaisemartin–D’Haultfœuille)

- more comparison units than PT-never, without restricting pre-trends;
- the **middle ground** the Practitioner’s Guide favours in its application;
- composition of the comparison changes over t — report it.

Their advice

“We strongly recommend that researchers clearly state the specific parallel trends assumption they are actually imposing, to allow readers to discuss its plausibility.”

Why estimates differ

Applied to the same data, two heterogeneity-robust estimators generally yield different estimates. Each is the sample counterpart of a different parallel-trends assumption, so a discrepancy points to the assumption on which they disagree:

- Imputation $\neq$ Sun–Abraham: the pre-period is not parallel, so the PT-all assumption is doing work, or the choice of base period matters.
- Callaway–Sant’Anna never $\neq$ not-yet: never-treated and not-yet-treated units trend differently; one of the two is a poor comparison group, and the choice has to be defended.
- Any of them $\neq$ TWFE: effects are dynamic or heterogeneous across cohorts, transmitted into the TWFE coefficient by the forbidden comparisons.

Do not run these estimators as a robustness check

They estimate different parameters under different assumptions: their agreement is not a confirmation, their disagreement not a failure. Choose the assumption you can defend, report the corresponding estimator, and use the others to show which assumption the result depends on.

Section 5

Aggregation

From building blocks to one number

ATT(g, t) are many, noisy, and cohort-specific. Aggregate them with **non-negative, transparent** weights $w_{g,t}$ that sum to one:

$$\text{ATT}^{\text{agg}} = \sum_{g,t} w_{g,t} \text{ATT}(g, t).$$

- **Event-time** aggregation: for each k , average ATT($g, g + k$) over the cohorts observed k periods after treatment (all states k years after their adoption), weighted by **cohort size** — the staggered event study.
- **Balanced** event time: keep only cohorts observed at every k in a window, so the composition does not change along the curve.
- **Calendar-time** and **by-cohort** aggregations; an **overall** ATT as the average of post-treatment event times.

Contrast

TWFE is also an aggregation of 2×2 s — with weights chosen by the variance of d_{it} , not by the question, and with forbidden comparisons inside.

Section 6

Workflow

Conclusion: the forward-engineering workflow (Baker et al. 2026)

- 1 **Define the target** — $ATT(g, t)$ and the aggregation you care about.
- 2 **State the identifying assumptions** — which parallel trends (never / not-yet / all), no-anticipation; argue them and bring pre-period evidence.
- 3 **Choose the estimator** that is the sample analogue of those assumptions.
- 4 **Discuss the source of uncertainty** — sampling (which units are observed) or design-based (which units are treated); cluster accordingly.
- 5 **Estimate.**
- 6 **Sensitivity analysis** — bounds on parallel-trends violations; the same target under the neighbouring assumption.
- 7 **Heterogeneity** — by cohort, by event time, by group.
- 8 **Keep learning** — if the assumptions are not credible, change design, not estimator.

On TWFE

“It has well-understood, potentially serious, and easily remedied problems, and we do not recommend using it.” — Baker et al. (2026)

Section 7

In R

What you must build in the data

These estimators need the **timing** structure explicitly:

- the **cohort** g_i — each unit's first treatment period, on *all* of its rows;
- **event time** $k = t - g_i$;
- a **never-treated** indicator (or the last-treated cohort as never-treated, dropping the periods where everyone is treated);
- units treated in the **first** period have no pre-period: drop them.

Software builds the 2×2 s and the aggregation for you — you supply g_i and the **assumption**.

In R: the timing variables (data.table)

Six lines build the variables the authors' R packages need, with `data.table`.

```
library(data.table); library(fixest); library(did)
library(did2s); library(HonestDiD)
d <- fread("rica_bio_td5.csv") # 1
d[, e := fifelse(cohorte > 0, annee - cohorte, NA_integer_)] # 2
d[, traite := as.integer(cohorte > 0 & annee >= cohorte)] # 3
d[, ee := e]; d[ee < -4 | ee > 5 | is.na(ee), ee := -1000L] # 4
d[, g_sa := fifelse(cohorte == 0, 10000L, cohorte)] # 5
d[, reg_an := paste(region, annee)] # 6
```

- 1 — one row per farm id and year `annee`; `cohorte` is g_i , the conversion year, 0 if never converted; y on the next slides stands for the outcome column.
- 2 — event time $k = t - g_i$, missing if never converted (`d[, x := ...]` adds a column; `fifelse` is a vectorised if).
- 3 — $d_{it} = \mathbf{1}\{t \geq g_i\}$, the treatment indicator of TWFE.
- 4 — event time **binned**: outside $[-4, 5]$ and never converted get a reference code.
- 5 — the cohort as `sunab` wants it: never converted coded as a far-future cohort.
- 6 — the region $\times$ year label of the fixed effect.

In R: TWFE and Sun–Abraham (fixest)

Four lines estimate the regression to avoid and its saturated correction.

```
tw <- feols(y ~ traite | id + reg_an, d, cluster = ~id)          # 1
sa <- feols(y ~ sunab(g_sa, annee, ref.p = -1) | id + reg_an,    # 2
            d, cluster = ~id)
summary(sa, agg = "ATT")                                         # 3
coeftable(sa)                                                    # 4
```

- 1 — TWFE: the outcome on d_{it} with farm and region $\times$ year effects, standard errors clustered by farm. Its coefficient is the weighted average of 2×2 s of the chapter, forbidden comparisons included.
- 2 — Sun–Abraham: `sunab(cohort, period)` builds every cohort $\times$ event-time dummy, with $k = -1$ as reference; the never converted are the only control.
- 3 — aggregates the cohort $\times$ time coefficients into one ATT with the cohort shares as weights.
- 4 — the event-study coefficients, one per k , with their standard errors.

In R: imputation (did2s)

Two lines estimate the PT-all analogue and test its pre-trends.

```
imp <- did2s(d, yname = "y", first_stage = ~ 0 | id + reg_an,  
             second_stage = ~ i(ee, ref = c(-1, -1000)),  
             treatment = "traite", cluster_var = "id")          # 1  
wald(imp, keep = "ee::-[2-4]$")                                # 2
```

- 1 — imputation in two stages. `first_stage`: the fixed effects are estimated on the **untreated cells only**, those where `traite` is 0. `second_stage`: the outcome net of those effects is regressed on the event-time dummies, `i(ee, ...)`, with $k = -1$ and the binned code as references. Standard errors clustered by farm.
- 2 — the joint test that the coefficients of $k = -4, -3, -2$ are zero. `fixest` names the dummies `ee::-4, ee::-3, ...`; `keep` selects them by a pattern on the name. This is the pre-trend check of PT-all.

In R: Callaway–Sant’Anna (did)

Two lines estimate every $ATT(g, t)$ and aggregate them by event time.

```
a <- att_gt(yname = "y", tname = "annee", idname = "id",  
            gname = "cohort", data = d,  
            control_group = "notyettreated",  
            base_period = "universal",  
            allow_unbalanced_panel = TRUE) # 1  
aggte(a, type = "dynamic", min_e = -4, max_e = 5) # 2
```

- 1 — one $ATT(g, t)$ per cohort and year, each a 2×2 against the **not-yet-treated**; "nevertreated" gives the other assumption. "universal" takes $g - 1$ as the base period for the pre-periods too (the default compares each pre-period with the previous one). On a rotating panel, `allow_unbalanced_panel` makes the package treat the data as repeated cross-sections.
- 2 — aggregates the $ATT(g, t)$ by event time, from -4 to 5 ; `type = "simple"` returns a single ATT instead.

In R: how much violation survives (HonestDiD)

Three lines bound the effect at one event time under a limited violation of parallel trends.

```
beta <- coef(imp); V <- vcov(imp) # 1
hd <- createSensitivityResults_relativeMagnitudes(
  betahat = beta, sigma = V, numPrePeriods = 3,
  numPostPeriods = 6, Mbarvec = c(0.5, 1, 1.5, 2),
  l_vec = basisVector(4, 6)) # 2
hd[, c("Mbar", "lb", "ub")] # 3
```

- 1 — the event-study coefficients of the imputation and their covariance, ordered from $k = -4$ to $k = 5$: three pre-periods $(-4, -3, -2)$ and six post $(0, \dots, 5)$; check the order with `names(beta)`.
- 2 — Rambachan–Roth: for each $\bar{M}$, the interval that stays valid if the post-treatment deviation from parallel trends is at most $\bar{M}$ times the largest pre-treatment one; `l_vec` selects the fourth post period, $k = 3$.
- 3 — one row per $\bar{M}$ with the bounds. The first $\bar{M}$ whose interval contains zero measures how thin the margin is.

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